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A knowledge-driven framework for predicting single-cell responses for unprofiled drugs

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Predicting cellular response to chemical perturbations is critical to build virtual cells, yet experimentally profiled compounds cover only a small fraction of this space. Existing models struggle to generalize to unprofiled compounds, as they typically treat drugs as isolated identifiers without encoding their mechanistic relationships. Here, we present MAP, a framework that integrates structured biological knowledge into cellular perturbation modelling and supports zero-shot prediction for small molecules with scarce or absent profiles. (1) We construct MAP-KG, a knowledge graph that unifies 14 public resources, spanning 187,089 drugs, 22,924 genes and 694,246 mechanistic relationships. (2) We propose a knowledge-driven pretraining strategy that aligns molecular structures, protein sequences and textual mechanistic descriptions into a unified embedding space, producing mechanism-aware and transferable gene and compound embeddings. These representations are then coupled with a pretrained single-cell foundation model to condition perturbation response prediction. (3) We evaluate MAP under two zero-shot generalization regimes: unseen cell type–drug combinations and a stricter setting of unprofiled drugs, where it improves the top-50 differentially expressed gene Pearson delta correlation by up to +12.3% and +11.8%, respectively, over the strongest baselines across three benchmarks. We further perform pathway-level functional analysis via gene set enrichment analysis for in silico screening, where MAP predicts mechanism-consistent programmes on unprofiled candidate drugs, and prioritizes four out of five approved anti-cancer drugs in A-549 (non-small-cell lung cancer).

Single-cell perturbation profiling is transforming our ability to interrogate causal drug actions by measuring transcriptome-wide responses at cellular resolution[1–5]. By linking chemical interventions to downstream gene programmes across heterogeneous cell states, these atlases provide a critical substrate for constructing virtual cells, that is, predictive models that forecast cellular responses under previously unobserved perturbations and contexts[6–8]. If accurate and generalizable, such models could enable scalable in silico screening across large chemical spaces, prioritizing candidate therapeutics and dosing strategies before physical experimentation and, consequently, reducing the cost and latency of early stage drug discovery[9].

Despite the rapid progress driven by larger perturbation atlases and more expressive models[10–16], robust generalization to unprofiled compounds remains a central challenge. In many atlas-scale settings, compounds are treated as categorical identifiers, and perturbation effects are inferred primarily from observed expression profiles[14,15].

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Such representations implicitly place drugs in a latent space in which proximity is learned only from co-occurrence in the training atlas, rather than from shared biological mechanisms, such as shared binding modes or similar pathway modulation. As a result, mechanistically related compounds may be encoded as unrelated tokens, limiting extrapolation when a test compound has little or no transcriptional profiling data.

A natural remedy is to condition predictors on drug descriptors such as molecular structure or targets, which can partially support out-of-distribution prediction. However, these descriptors are heterogeneous and incomplete in practice, and they do not, by themselves, provide a unified interface for integrating diverse biological evidence (for example, simplified molecular input line entry system (SMILES) strings, protein sequences, pathway membership and free-text mechanism descriptions) into a single mechanism-aware representation. Consequently, current models[10,12,17–21] still struggle to systematically transfer such knowledge to the setting that matters most for discovery: predicting transcriptional responses for compounds outside of the profiled atlas[22].

Here we present a mechanism-aware perturbation response predictor (MAP), a framework that integrates structured pharmacological knowledge into cellular response prediction. MAP learns mechanism-aware drug and gene representations by aligning molecular structures, protein targets and mechanistic descriptions in a unified embedding space, and then conditions a perturbation predictor on these knowledge-informed representations. This design enables extrapolation to compounds with scarce or absent perturbation profiles, providing a mechanistically grounded route towards in silico screening at scale.

Specifically, we make the following contributions:

On knowledge injection, we start by constructing MAP-KG (Fig. 1a), a perturbation-oriented biomedical knowledge graph that consolidates evidence from 14 public resources. MAP-KG links 187,089 drugs and 22,924 genes through 694,246 mechanistic relations, and leverages text (for example, mechanisms of action (MOA) and functional descriptions) as a semantic bridge connecting heterogeneous attributes such as SMILES strings and protein sequences. Building on MAP-KG, we develop modality-specific encoders and align them via contrastive learning to map multisource evidence into a shared embedding space (Fig. 1b).

On model development, we propose a knowledge-driven pretraining strategy that learns transferable drug and gene representations from MAP-KG by jointly aligning multimodal attributes and enforcing relation-level consistency. These representations provide a mechanistic prior that moves beyond treating compounds as independent identifiers. We integrate the resulting knowledge encoders into a perturbation predictor built on a single-cell foundation model, combining control-state cell embeddings with knowledge-informed drug and gene embeddings to forecast perturbed transcriptomes (Fig. 1c).

On experiment evaluation, we validate MAP along multiple axes. (1) For zero-shot compositional generalization to unseen cell type–drug combinations, MAP improves the top-50 differentially expressed gene (DEG) Pearson delta correlation by 12.3% on Tahoe-100M and 6.6% on SciPlex3, and improves direction accuracy by 6.7% on the OP3 benchmark. (2) For zero-shot prediction of unprofiled drugs (no perturbation profiles observed during training), MAP improves the top-50 DEG Pearson delta correlation by 11.8% (Tahoe-100M), 19.8% (OP3) and 10.4% (SciPlex3), and further achieves consistent improvements in two-drug-combination perturbation prediction across increasingly stringent drug combination splits. (3) Using pathway-level gene set enrichment analysis (GSEA) read-outs, MAP recovers coherent, mechanism-consistent programmes and supports in silico screening in A-549 (non-small-cell lung cancer (NSCLC)), prioritizing four out of five approved anti-cancer drugs within the top 15 among 58 held-out compounds. (4) Finally, through progressive experiments, we show that scaling mechanistic knowledge provides a complementary axis to data and model capacity, yielding additional gains in zero-shot generalization to unprofiled compounds with substantially lower resource costs.

Results

We present MAP, a mechanism-aware predictor of transcriptional responses to chemical perturbations that leverages structured biomedical evidence to generalize to compounds with limited or no profiling data. We evaluate MAP in two zero-shot regimes: (1) unseen cell line–drug combinations and (2) entirely unprofiled drugs, and assess performance using both gene-level metrics and pathway-level GSEA read-outs to test functional coherence and utility for in silico screening.

Benchmark description

Evaluation settings. We evaluate MAP under two zero-shot protocols of increasing stringency.

(1) Zero-shot compositional generalization (unseen cell line–drug combinations). This setting probes context transfer in which a drug may be observed in the atlas but not in a particular cellular context. We hold out a subset of cell line–drug pairs during training and evaluate performance on these unseen combinations.

(2) Zero-shot prediction for unprofiled drugs (unseen compounds). This setting mimics virtual screening for candidates with no perturbational measurements. We remove all perturbation profiles of a subset of drugs from training and evaluate on these held-out compounds. To prevent information leakage, we also exclude the held-out drugs and relevant relations from MAP-KG during knowledge-driven pretraining; therefore, inference for unprofiled compounds relies only on their available attributes (for example, molecular structure and associated annotations) rather than any atlas-derived transcriptional evidence.

Datasets. We benchmark MAP across four large-scale datasets covering distinct biological contexts. (1) Tahoe-100M[23], a giga-scale atlas spanning 50 cancer cell lines and 379 drugs. To provide a proof-of-concept evaluation and keeping computation tractable, we restrict training and validation to six cell lines, spanning diverse tissues and cancer types. (2) OP3 (ref. 24), covering 144 compounds across six immune cell types. (3) SciPlex3 (ref. 2), profiling 187 compounds across three cancer cell lines. Together, these datasets span diverse cellular contexts and perturbation libraries, enabling the systematic evaluation of drug generalization from large cancer atlases to immune benchmarks. We additionally use ComboSciPlex[25] for an auxiliary evaluation of two-drug-combination perturbation prediction in which MAP is trained on single-drug perturbations and tested on held-out drug combinations.

Metrics. We evaluate response prediction using five complementary metrics: Pearson delta correlation for measuring concordance in perturbation-induced expression changes, direction accuracy for assessing agreement in up- or downregulation, perturbation discrimination score for evaluating separability across perturbations, mean square error for quantifying pointwise prediction error, and Wasserstein distance for measuring distributional discrepancy. Formal definitions are provided in Supplementary Section A.9. To mitigate single-cell measurement noise, all evaluations are performed at pseudobulk resolution by randomly aggregating a fixed number of cells from the same condition, following ref. 15. We compute the above metrics on two canonical gene sets: top-2,000 highly variable genes (HVGs) and top-50 DEGs. The perturbation discrimination score is calculated only on HVG to ensure the consistency of gene sets across perturbations.

Baseline models. We compare MAP against state-of-the-art perturbation prediction baselines, including models designed for compositional generalization across contexts and models that condition on molecular structure. Specifically, we include CRISP[19], a single-cell-foundation-model-assisted model for zero-shot prediction

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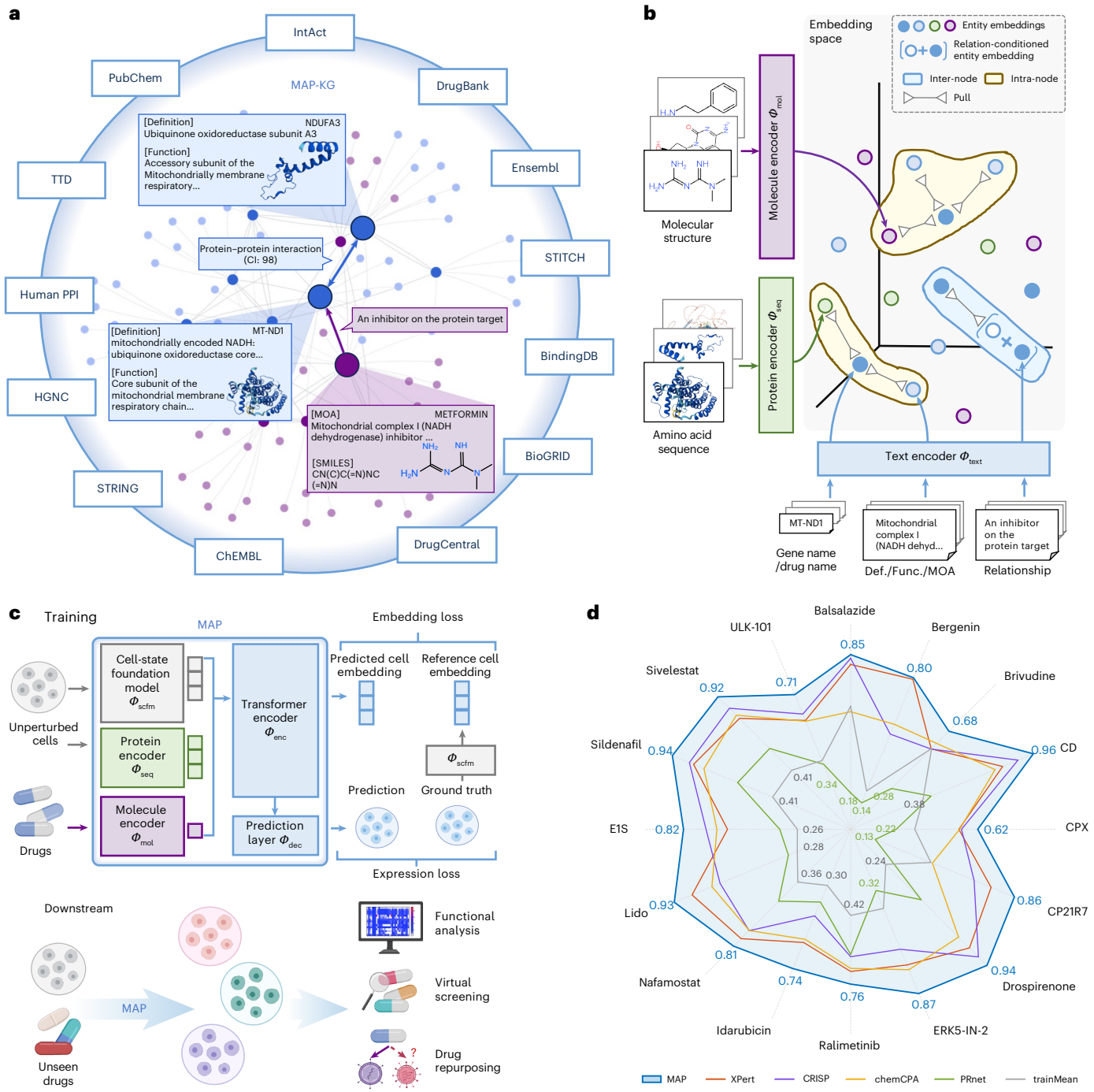


Fig. 1 | Overview of MAP-KG and MAP. a, We integrate 13 authoritative public databases shown in the figure, together with PrimeKG, a pre-existing biomedical knowledge graph that aggregates knowledge from multiple biological databases, to construct a large-scale biomedical knowledge graph, comprising 187,089 drugs, 22,924 genes and 694,246 triplets. **b**, We adopt modality-specific encoders for knowledge encoding, and conduct knowledge-driven pretraining to align drug and genes entities with biological mechanism in the representation geometry. **c**, MAP integrates the knowledge-informed drug and gene representation with a single-cell foundation model to guide perturbation response prediction. Consequently, MAP can generalize to unprofiled drugs without any transcriptional perturbation data, facilitating applications like functional analysis, virtual screening and drug repurposing. **d**, MAP demonstrates state-of-the-art predictive performance across a variety of unprofiled drugs. E1S, estrone sulfate; CPX, ciclopirox; CD, carbidopa; Lido, lidocaine; Ralimetinib, ralimetinib dimesylate. Illustrations in **b** and **c** created in BioRender; Zhou, X. https://biorender.com/rxukvvg (2026).

on unseen cell type–perturbation combinations; chemCPA[10], a compositional perturbation autoencoder that injects drug representations into a latent basal cell state; PRnet[18], a deep generative model that conditions on molecular structure and control-state expression to predict responses to unseen compounds; and XPert[21], a dual-branch transformer that integrates graph-derived biological tokens with chemical features. We also include a competitive linear baseline, trainMean[22,26], which predicts perturbation effects using mean responses estimated from the training set. We also compare against STATE[15] for zero-shot compositional generalization, excluding it from the unprofiled-drug regime because its fixed categorical perturbation labels preclude inference on unseen drugs.

Zero-shot generalization across unseen cell line–drug combinations

We first assess zero-shot compositional generalization to unseen cell line–drug combinations. In this setting (Fig. 2a), we split at the pair level: for each cell line, we hold out 5% of drugs only in that cell line for testing, and ensuring that the held-out drug sets are non-overlapping across cell lines. Consequently, every test drug remains observed in the training set in other cellular contexts, but its response in the held-out cell line must be inferred without paired supervision. This protocol reflects a repurposing-like scenario in which a compound profiled in some contexts is evaluated in a previously unobserved cellular setting.

Overall performance on Tahoe-100M. As shown in Fig. 2b, on Tahoe-100M, MAP substantially outperforms all baselines across metrics, achieving relative improvements over the best-performing baseline of 11.3% in the top-50 DEG direction accuracy (STATE), 12.3% in top-50 DEG Pearson delta correlation (STATE) and 15.2% in the HVG perturbation discrimination score (CRISP). Details are provided in Supplementary Table 1. These gains indicate that MAP more faithfully matches both the magnitude of perturbation effects (higher Pearson delta correlation) and the direction of regulation (higher direction accuracy), as well as produces more separable perturbation signatures in the predicted space (higher HVG discrimination score). MAP also achieves the lowest MSE and Wasserstein distance, indicating more accurate predictions and better distributional alignment (Supplementary Table 1). Together, the results suggest that mechanism-informed compound representations improve the transfer of perturbational effects across cellular contexts.

Cell-line-specific performance on Tahoe-100M. We next stratify the performance by cell line (Fig. 2c). Across six diverse cancer cell lines spanning multiple tissues and disease contexts, MAP achieves consistently strong results: top-50 DEG Pearson delta correlation reaches ≥0.95 on A-172 and remains close to 0.90 on the lowest-performing Hep-G2/C3A line. Direction accuracy is similarly high and stable across cell lines, indicating a robust recovery of gene-wise up- and downregulation. CRISP and XPert also improve substantially over the linear train-Mean baseline, but their gains are smaller and exhibit greater variability across cell lines than MAP (Supplementary Table 2).

Generalization to unseen cell line–drug combinations across datasets. To evaluate robustness beyond Tahoe-100M, we apply the same unseen-combination protocol to SciPlex3 and OP3, which contain 28 and 42 non-overlapping cell type–drug combinations, respectively, using the same training and evaluation setup as in Tahoe-100M. As shown in Fig. 2d, MAP consistently improves the top-50 DEG direction accuracy, top-50 DEG Pearson delta correlation and HVG perturbation discrimination score on both datasets, indicating reliable zero-shot inference across experimental settings and extending from cancer cell lines to primary immune cell types. We note that HVG perturbation discrimination score values are relatively low for all methods on SciPlex3 and OP3, which may reflect weaker perturbation separability arising from dataset-specific protocol differences and experimental configurations.

Cell-type-specific analysis on SciPlex3 and OP3. Finally, we examine cell-type-specific performance (Fig. 2e). On SciPlex3, MAP is consistently strong across the three cell lines, achieving a top-50 DEG Pearson delta correlation of 0.857 on K562. On OP3, MAP similarly maintains strong performance across all six immune cell types; even in myeloid cells, where correlations are lower for all models, MAP reaches 0.730, indicating meaningful agreement between the predicted and observed perturbation effects (Supplementary Table 3).

Zero-shot generalization to unprofiled drugs

We next evaluate MAP in a more challenging regime: zero-shot prediction for unprofiled drugs, where test compounds are entirely absent from perturbation training data (Fig. 3a). This setting probes whether the model can infer perturbation effects from molecular attributes (for example, chemical structure), without access to any transcriptional response measurements for the test compounds in any cell type. We hold out 5% of drugs as the test set and remove all the corresponding perturbation profiles from training. To avoid information leakage, we additionally exclude the held-out drug entities and their aliases from MAP-KG during knowledge pretraining. Edges involving the held-out drugs were also removed to prevent indirect information leakage.

Overall performance on Tahoe-100M. As shown in Fig. 3b, MAP substantially improves over baseline methods across all metrics in the unprofiled-drug regime, with +17.5% top-50 DEG direction accuracy, +11.8% top-50 DEG Pearson delta correlation and +16.9% HVG perturbation discrimination score. MAP attains the lowest MSE and Wasserstein distance, reflecting higher prediction accuracy and closer alignment with the target distribution (Supplementary Table 5). In particular, MAP also improves performance on seen drugs, suggesting that knowledge injection not only supports extrapolation to unseen compounds but also strengthens in-distribution modelling by providing a more informative mechanistic prior (Supplementary Tables 4 and 5).

Per-compound performance. We further analyse the results at the level of individual held-out compounds (Fig. 3c). MAP achieves the highest top-50 DEG Pearson delta correlation on all of 16 test compounds. Across compounds, performance ranges from 0.942 on sildenafil to 0.620 on ciclopirox, indicating robust zero-shot prediction despite diverse chemical structures and heterogeneous mechanisms (Supplementary Tables 6 and 7).

Gene-level agreement for representative drugs. To examine gene-level behaviour, we visualize single-gene predictions for idarubicin (hydrochloride) in A-172 and A-549, comparing MAP with CRISP and XPert (Fig. 3d). MAP shows tighter alignment with the diagonal, indicating closer agreement between the predicted and observed gene-wise log[fold change] values. Consistently, the Pearson delta correlation on HVG is higher for MAP: on A-172, 0.685 versus 0.447 (CRISP) and 0.536 (XPert); on A-549, 0.743 versus 0.489 and 0.513, respectively. Although a small number of outliers remain, MAP better captures the perturbation magnitudes with reduced dispersion and fewer systematic biases than the baselines. These results demonstrate that MAP more faithfully recovers fine-grained transcriptional perturbation patterns at the single-gene level, highlighting a superior ability to model expression change across different cellular contexts.

Generalization to unprofiled drugs across datasets. Finally, we evaluate unprofiled-drug generalization on SciPlex3 and the OP3 benchmark (Fig. 3e). MAP maintains strong gene-level accuracy, achieving an average top-50 DEG Pearson delta correlation of 0.827 across SciPlex3 cell lines and 0.861 across OP3 immune cell types, with consistent gains in direction accuracy and HVG perturbation discrimination score (Supplementary Tables 8 and 9). Together, these results support the robustness of MAP for zero-shot prediction on unprofiled compounds across datasets and heterogeneous cellular contexts.

Extension to drug combination perturbations. We further extended MAP to combinatorial drug responses, a practically relevant setting for modelling combination therapies. On the ComboSciPlex dataset, we trained MAP using only single-drug perturbations and evaluated it on two-drug combinations under three splits with decreasing drug observability: both drugs seen, one drug seen and neither drug seen during training. As reported in Supplementary Table 13, MAP consistently

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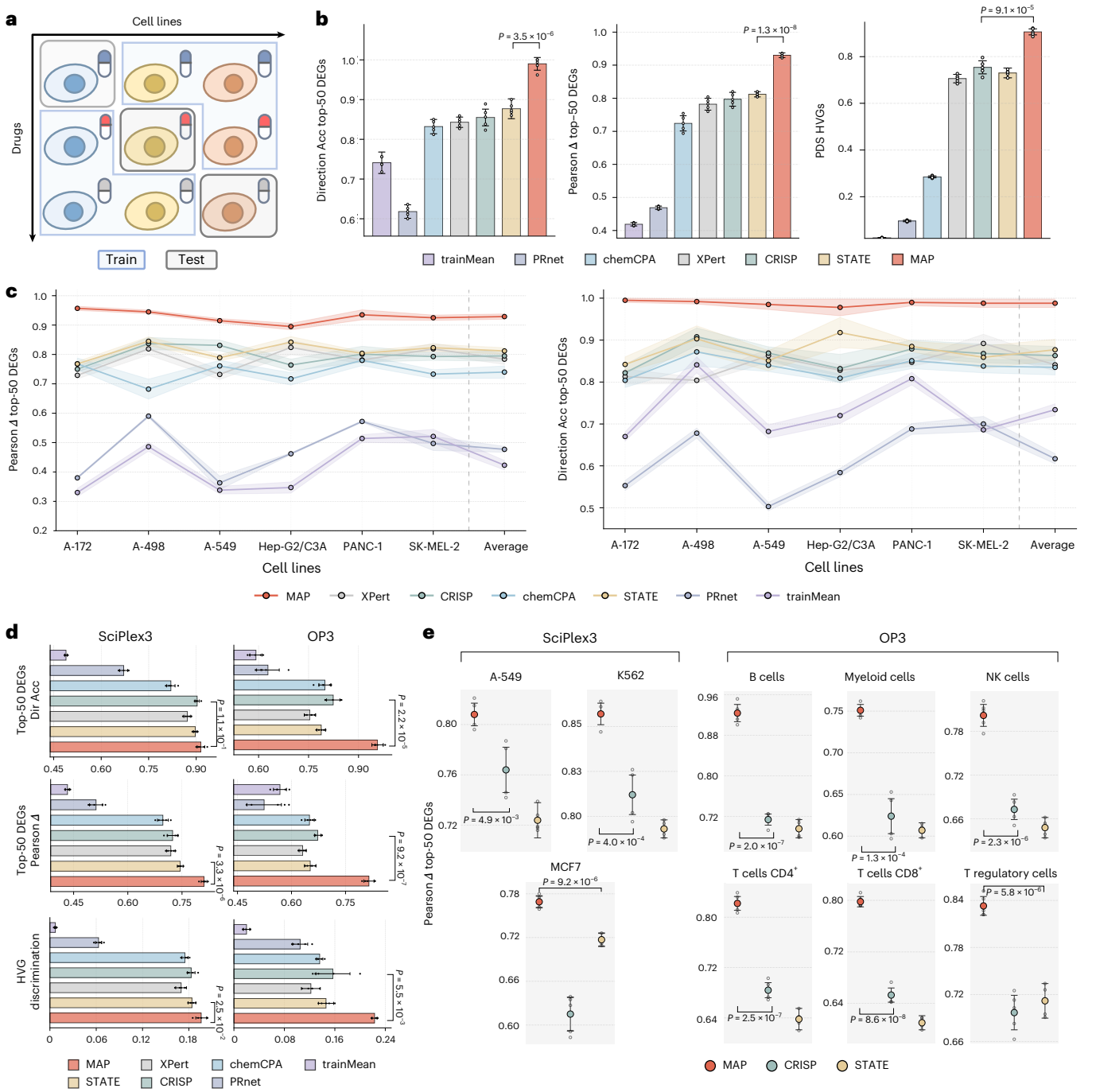


Fig. 2 | Zero-shot generalization performance on unseen cell line–drug combinations. a, Illustration of zero-shot generalization to unseen cell type–drug pairings. For each cell type, we hold out 5% of the drugs as unseen, and the held-out drug sets are non-overlapping across cell types. **b**, Performance comparison between MAP and baselines on six cell lines selected from the Tahoe-100M dataset. Three key evaluation metrics are shown. PDS, perturbation discrimination score. **c**, Cell-type-specific performance on Tahoe-100M. We report the top-50 DEG Pearson delta correlation and direction accuracy; shaded bands indicate the 95% confidence intervals (CIs). **d**, Performance comparison between MAP and baselines on the SciPlex3 and OP3 datasets. Three key evaluation metrics are shown. Dir Acc, direction accuracy. **e**, Cell-type-specific performance on the SciPlex3 and OP3 datasets. For each cell type, we present the results of the top-3 performing methods. For bar charts, data are presented as mean values with 95% CIs across $n = 5$ independent evaluations on the same held-out test set. The P values were computed using two-sided Welch's t-tests between MAP and the second-best model for each metric, without adjustment for multiple comparisons. The whiskers represent the 95% CI. Five independent data points are shown.

outperforms the Control and PerturbMean baselines across all settings and evaluation metrics. The performance degradation from seen to fully unseen drug pairs is also generally more gradual for MAP, suggesting that its mechanism-aware drug representations provide useful priors for combinatorial perturbation generalization. Detailed experimental settings and analysis are provided in Supplementary Section A.2.

Functionally interpretable pathway responses

To improve functional interpretability, we evaluate pathway-level responses by applying GSEA[27] to model-predicted perturbation profiles. This shifts assessment from individual genes to coordinated biological programmes that are more directly actionable in biomedical settings. We focus on the A-549 cell line, a widely adopted

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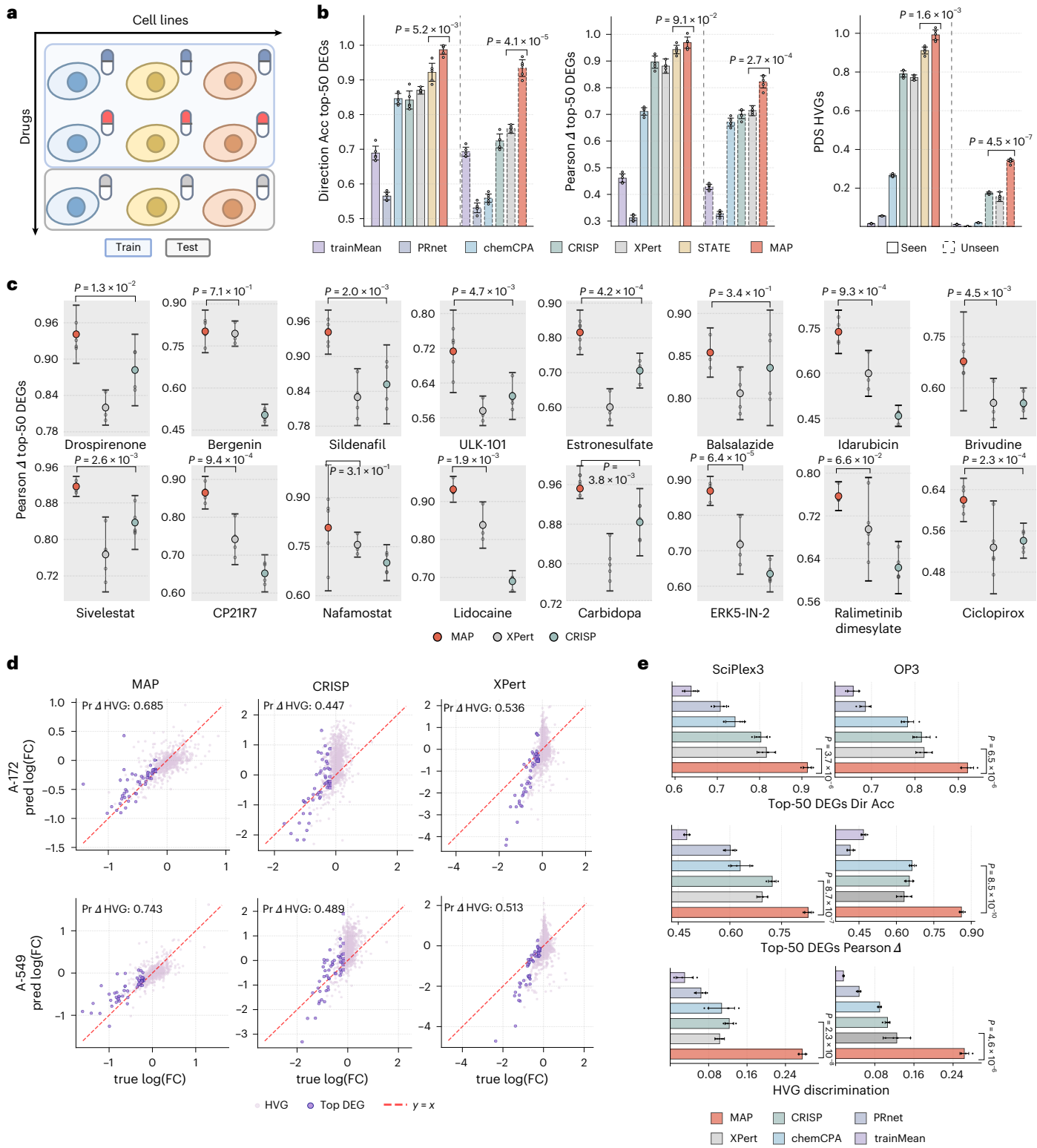


Fig. 3 | Unprofiled drug generalization performance of MAP. a, Illustration of the drug generalization scenario, with 5% drugs withheld during training as unseen testing drugs. **b**, Average performance comparison between MAP and baseline models on six cell lines selected from the Tahoe-100M dataset. Three evaluation metrics are shown. **c**, Compound-specific analysis of top-50 DEG Pearson delta correlation, where 95% CI and five data points are shown. **d**, Visualization of the single-gene-level perturbation predictions for idarubicin (hydrochloride) on A-172 and A-549 cell lines. The top-50 DEGs and top-2,000 HVGs are marked with distinct colours. Pearson delta correlations of the HVGs are annotated in each panel. true log(FC), true value of log(fold change); pred log(FC), predicted value of log(fold change). **e**, Performance comparison between MAP and baseline models on the OP3 and SciPlex3 datasets. Three evaluate metrics are demonstrated. For bar charts, data are presented as mean values with 95% CIs across $n = 5$ independent evaluations on the same held-out test set. The P values were computed using two-sided Welch's t-tests between MAP and the second-best model for each metric, without adjustment for multiple comparisons. The whiskers represent 95% CI, whereas five independent data points are shown using small dots.

model of lung adenocarcinoma and NSCLC drug response. We curate an expanded set of 58 candidate drugs from Tahoe-100M, including five drugs that have been approved for use in A-549-related contexts: afatinib, osimertinib, crizotinib, adagrasib and pemetrexed. Following the unprofiled-drug protocol (see the 'Zero-shot generalization to unprofiled drugs' section), we train MAP and perform zero-shot prediction on the held-out candidate compounds.

Simulated in silico drug screening. We conduct a simulated screening experiment on the A-549 cell line. We curate a set of disease-relevant pathways, including major oncogenic drivers and core signalling axes implicated in lung adenocarcinoma progression. Following ref. 19, for each drug, we compute pathway enrichment scores from the predicted transcriptional response and aggregate them across the curated pathways to obtain a drug-level downregulation score, where lower scores indicate stronger predicted suppression and, therefore, more promising candidates. Figure 4a reports drug ranking by the inverse downregulation score. Four clinically approved lung cancer drugs[28–33] appear among the top-14 candidates; in particular, adagrasib and afatinib rank second and fourth among all held-out drugs, respectively. Beyond approved drugs, literature mining of the top-ranked non-standard candidates identified homoharringtonine as a literature-supported candidate with direct A-549 evidence, including the reported inhibition of A-549 proliferation, clonogenic growth and cancer-relevant signalling programmes such as STAT3 signalling and TMEM16A-related channel activity, supporting the ability of MAP to nominate biologically meaningful candidates beyond simply retrieving approved therapies[34–36]. A detailed analysis is provided in Supplementary Section A.7. Finally, we quantify retrieval performance by computing the top-k recall of the approved drugs and compare against CRISP and chemCPA. As shown in Fig. 4b, MAP achieves consistently higher recall across k, suggesting improved virtual screening utility.

Gene-set-level mechanism agreement. To assess whether MAP captures biologically meaningful programmes, we perform a fine-grained analysis of pathway regulation. Figure 4c summarizes the GSEA results for the four approved drugs that rank the highest during screening. Across these four drugs, the predicted normalized enrichment scores (NES) closely match the ground-truth NES for the highlighted gene sets, correctly recovering both direction and relative magnitude of pathway regulation. These results indicate that MAP can produce functionally interpretable perturbation predictions at the pathway level that are consistent with established biological mechanisms. For each drug, we additionally examine representative pathways linked to its known MOA and to disease-relevant signalling in A-549 cells (Supplementary Sections A.6 and A.8 and Supplementary Table 10).

Biological knowledge enhances zero-shot generalization to unprofiled drugs

We posit that structured mechanistic knowledge serves as a critical inductive bias, enabling the model to extrapolate perturbation effects to compounds with limited or absent transcriptional supervision. We validate and dissect the contributions of biological knowledge in unprofiled-drug generalization (see the 'Zero-shot generalization to unprofiled drugs' section), along two complementary axes: (1) on the knowledge scale, we perform progressive knowledge injection by sequentially incorporating knowledge sources (Fig. 5a); (2) on knowledge diversity, we conduct category-wise ablation, selectively withholding specific knowledge categories (Fig. 5b). In addition, we visualize drug and gene entities in the latent space (Fig. 5c) to demonstrate that knowledge enhancement yields mechanism- and annotation-consistent representation geometry.

Generalization performance scales with the biological priors. We progressively increase the knowledge data in pretraining, divided into four stages. (1) No knowledge: we bypass the knowledge-driven pretraining and use randomly initialized drug embeddings. (2) MoleculeSTM: we skip pretraining and derive drug embeddings from MoleculeSTM[37], an off-the-shelf molecule encoder that provides physicochemical knowledge of chemical structures. (3) +PrimeKG: we adapt MoleculeSTM with the proposed knowledge-driven pretraining on PrimeKG[38], a knowledge graph centred on drug–disease interactions, containing mechanism-aware knowledge on 7,957 drugs. (4) +MAP-KG: we continue the pretraining on MAP-KG, a perturbation-oriented knowledge graph of an unprecedented scale. As shown in Fig. 5a, all three metrics improve monotonically as more knowledge data are included. In particular, adding MoleculeSTM improves the top-50 DEG direction accuracy and Pearson delta correlation by 27.8% and 37.7%, respectively, indicating that drug-side knowledge alone yields rapid gains. Using the full MAP-KG for pretraining further improves over PrimeKG-only pretraining, with additional gains of 7.2%, 11.4% and 4.3% on the top-50 DEG direction accuracy, Pearson delta correlation and HVG perturbation discrimination score, respectively, underscoring the benefit of scaling knowledge data for generalization performance (Supplementary Table 11).

Synergistic knowledge promotes generalization. To investigate the contribution of different biological knowledge, we ablate one knowledge category from MAP-KG at a time, including drug-side textual attributes (for example, MOA and therapeutic functions), gene-side attributes (for example, amino acid sequence and textual description of functional annotations) and relational edges (drug–gene mechanistic interactions and gene–gene functional and physical associations). As illustrated in Fig. 5b, removing any single category reduces the generalization performance, indicating that the gains are supported by multiple, complementary biological knowledge. In particular, removing drug–gene edges causes the largest drop in both top-50 DEG direction accuracy and Pearson delta correlation, highlighting the central role of drug–gene mechanistic interactions in cellular perturbation modelling (Supplementary Table 12).

Knowledge pretraining yields mechanism-aligned representations. We visualize drug and gene embeddings with uniform manifold approximation and projection (UMAP) to analyse how multimodal knowledge pretraining reshapes the geometry of entity representations (Fig. 5c). We quantify clustering quality using the Davies–Bouldin index (DBI; a lower value indicates tighter, better-separated clusters) and silhouette score (a higher value indicates better intra-cluster cohesion and inter-cluster separation). For drugs, the pretrained embeddings show clearer grouping by MOA than the raw MoleculeSTM features, with mechanistically similar compounds forming compact clusters (lower DBI and higher silhouette scores), indicating capturing transferable priors among drugs of related MOA. Similarly, for genes, pretrained embeddings exhibit sharper separation by gene family and domain annotations than the ESM-2 embeddings, producing more coherent gene clusters. Together, these patterns suggest that knowledge pretraining promotes mechanism-level information exchange between drug and gene entities, providing a strong basis for transferable, zero-shot perturbation prediction.

This work provides a unified framework that bridges mechanism-aware biological knowledge with high-dimensional single-cell readouts to enhance generalizable perturbation modelling. Specifically, we contribute MAP-KG (a large-scale perturbation-oriented knowledge graph that harmonizes heterogeneous biological knowledge) and MAP (a holistic methodology that transforms these priors into unified drug and gene embeddings and synergizes with existing cell-state foundation models). Below, we discuss the implications of our findings from three perspectives.

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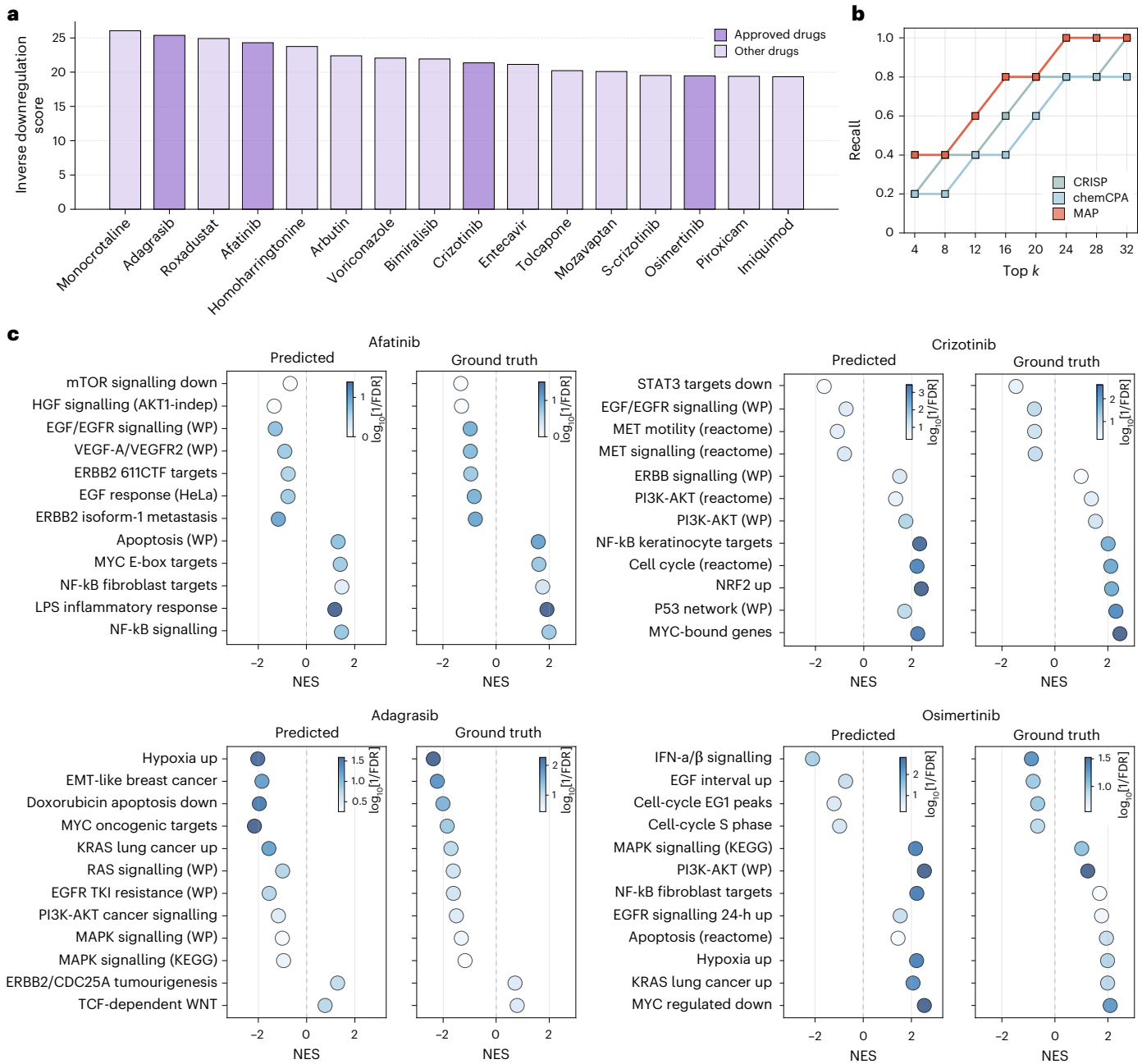


Fig. 4 | Functionally interpretable pathway responses. a, Virtual drug screening based on aggregated GSEA enrichment scores over curated A-549-relevant pathways, demonstrating top-16 drugs out of the 58 unseen drugs. Four clinically approved drugs are prioritized among unseen compounds. **b**, Top-*k* recall comparison of clinically approved drugs between MAP, CRISP and chemCPA, where MAP achieves higher recall at smaller *k*, indicating more efficient screening. **c**, Comparison of the predicted and experimental GSEA NES for key pathways across representative approved drugs. For each drug, 12 gene sets closely related to the A-549-disease-driving context and the drug's MOA are shown, and MAP closely matches the direction and relative magnitude of pathway regulation observed experimentally. HGF, hepatocyte growth factor; EGF, epidermal growth factor; EGFR, epidermal growth factor receptor; WP, WikiPathways; EMT, epithelial-to-mesenchymal transition; KEGG, Kyoto Encyclopedia of Genes and Genomes; TCF, T cell factor; WNT, wingless-related integration site.

Advancing robust and generalizable artificial intelligence virtual cells

Building robust artificial intelligence virtual cells hinges on the ability to accurately model cellular responses to perturbations. However, a fundamental asymmetry persists: the vastness of the chemical intervention space far outstrips the capacity of experimentally profiling. Despite recent progress, generalization to unprofiled or sparsely observed compounds remains a key bottleneck for existing data-driven paradigms. By introducing comprehensive biological priors, MAP achieves substantial advantages in zero-shot perturbation response prediction under unseen cell line–drug combinations (see the 'Zero-shot generalization across unseen cell line–drug combinations' section) and unprofiled drugs (see the 'Zero-shot generalization to unprofiled drugs' section). Crucially, the performance superiority is consistently observed across datasets and diverse cellular contexts, confirming that biological knowledge successfully serves as transferable priors even when transcriptional supervision is sparse or absent. Together, these results position knowledge-informed modelling as an important

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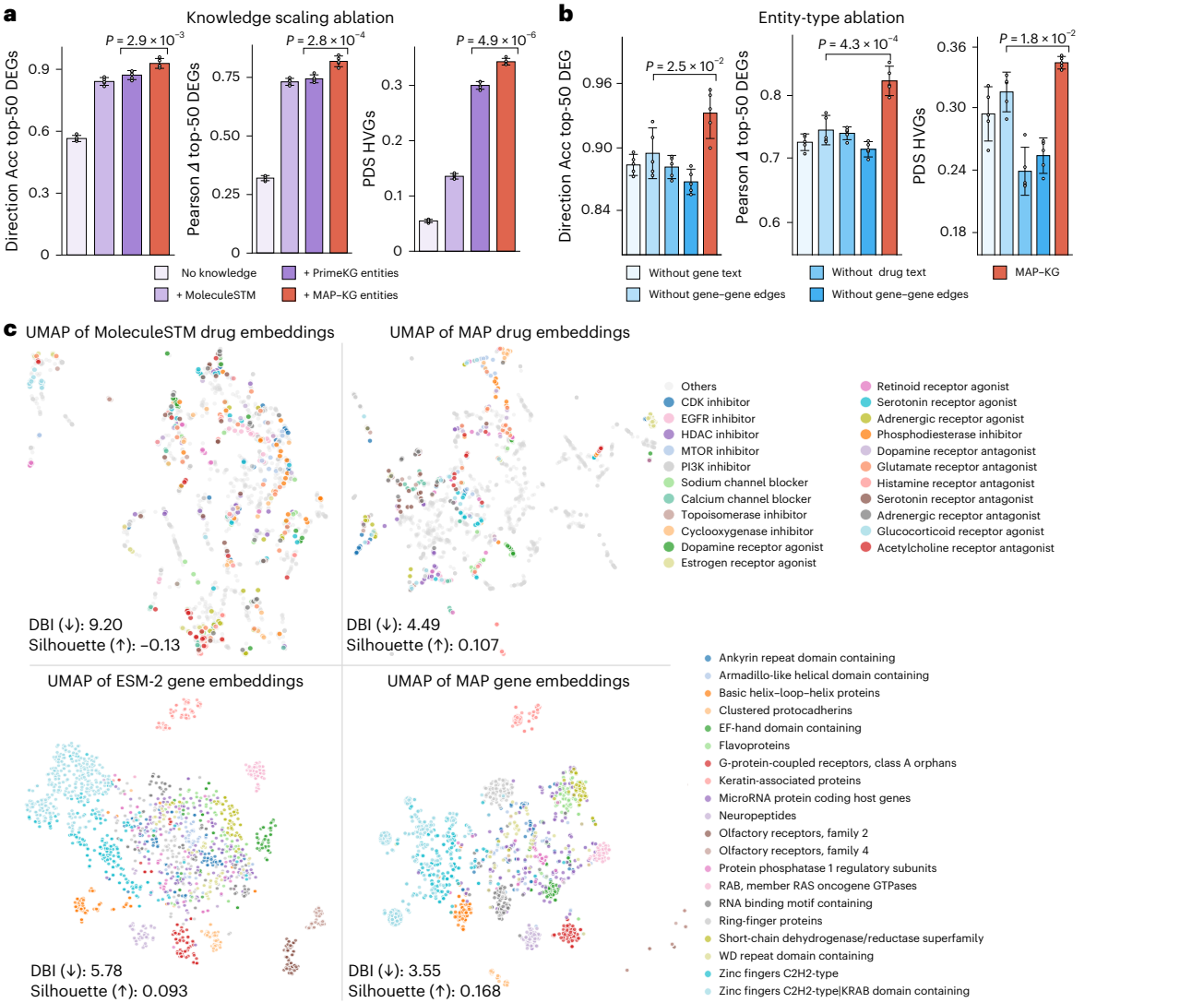


Fig. 5 | Ablations of knowledge scaling and visualization of knowledge representations. a, Performance on the unprofiled-drug regime as we progressively increase the injected knowledge sources. Knowledge variants are defined as follows: no knowledge, randomly initialized knowledge encoders; +MoleculeSTM, initialization from a pretrained MoleculeSTM checkpoint; +PrimeKG, further pretraining with PrimeKG entities; +MAP-KG, additional pretraining with MAP-KG entities. **b**, Ablation of knowledge types by removing one type of entities from MAP-KG at a time. We report the same three metrics under the unprofiled-drug protocol, highlighting the contribution of each knowledge type. **c**, UMAP visualization of drug embeddings (top) and gene embeddings (bottom) before and after knowledge pretraining. Points are coloured by MOA categories for drugs and by gene families for genes, showing clearer mechanism- and function-consistent organization after training. DBI and silhouette score are reported. For bar charts, data are presented as mean values with 95% CIs across $n = 5$ independent evaluations on the same held-out test set. The P values were computed using two-sided Welch's t-tests between MAP and the second-best model for each metric, without adjustment for multiple comparisons. The whiskers represent 95% CI, whereas five independent data points are shown by small dots. EF-hand, a calcium-binding helix–loop–helix motif; WD, tryptophan–aspartic acid; KRAB, Krüppel-associated box.

contribution to advancing robust and generalizable artificial intelligence virtual cells.

Accelerating therapeutic discovery via biologically faithful virtual screening

The high attrition rate and substantial costs of drug discovery necessitate effective in silico prioritization strategies. However, virtual screening depends not merely on numerical accuracy at the gene level but on the faithful prediction of biological functions and regulatory programmes. Our simulated screening on the A-549 lung cancer model highlights the translational potential of the proposed approach (see the 'Functionally interpretable pathway responses' section). In a strict zero-shot setting, MAP successfully prioritized clinically approved therapies, for example, adagrasib and afatinib, from a pool of unprofiled candidates, driven by the accurate prediction of disease-relevant pathway suppression. By delivering functional interpretable predictions, it could facilitate compound prioritization and drug repurposing, aligning computational inference with biological validation.

Knowledge enhancement as an orthogonal and cost-efficient paradigm

Despite the rapid expansion of perturbation atlases and the emergence of single-cell foundation models, data-driven approaches for cellular perturbation modelling still face substantial challenges, including the combinatorial complexity of the perturbation space, heterogeneous cellular contexts and prohibitive experimental costs. Our findings highlight the value of leveraging established biological knowledge as an informative and cost-efficient prior. We observe consistent improvement in zero-shot performance as the scale of the knowledge graph expands (see the 'Biological knowledge enhances

zero-shot generalization to unprofiled drugs' section). Furthermore, diverse biological knowledge, ranging from molecular structures to pathway interactions, collectively contributes to the model's generalization performance. This trend implies that mining broader and richer biological priors, including unstructured scientific literature, presents a direction for further improvement. Crucially, our approach is orthogonal to the underlying cell-state foundation model. Therefore, it offers a complementary axis alongside data and model scaling for improving perturbation modelling performance.

Future work

We identify two promising avenues to further advance this research. First, although this study focuses on small-molecule drugs, the proposed knowledge-driven pretraining can be extended to a broader perturbation landscape, since it captures general biological priors between genes, chemical compounds and pathways. Future work could adapt our approach to model genetic interventions or cytokine treatments, as relevant datasets are becoming available. Second, the rapid evolution of large-scale cell-state foundation models offers a powerful basis for modelling cellular heterogeneity and provides a promising substrate for incorporating mechanistic knowledge. A practical limitation of MAP is its relatively high computational and memory cost, which mainly arises from the gene-level tokenization used to model fine-grained interactions among cell states, drugs and genes. Developing more efficient architectures, such as optimized attention computation for long gene-token sequences, will be important for scaling knowledge-guided perturbation modelling to larger atlases and broader foundation-model settings.

Methods

MAP is a knowledge-enhanced framework for predicting cellular transcriptional responses to small-molecule perturbations. It addresses two key challenges: (1) generalizing to unseen cell type–drug combinations, and (2) zero-shot prediction for drugs absent from training data. The core idea is to leverage structured biomedical knowledge to learn drug and gene representations that capture mechanistic similarities, enabling robust generalization across compounds and cellular contexts.

Problem formulation

Given a small-molecule perturbation $p \in \mathcal{P}$, our goal is to predict the cellular transcriptional response under a specific cell context. Formally, let $x \in \mathbb{R}^G$ denote the gene expression profile of a cell over G genes under the control condition. Within a fixed cell context, we aim to learn a function that predicts the corresponding post-perturbation expression $\hat{x}^{(p)}$:

$$f_\theta : (x, p) \to \hat{x}^{(p)}. \tag{1}$$

In practice, to reduce measurement noise, we perform bulk-of-cells prediction (B unperturbed cells $X \in \mathbb{R}^{B \times G}$) and take the average.

We evaluate generalization in two out-of-distribution settings critical for drug discovery. (1) Unseen cell type–drug combinations: predicting responses for unseen combinations in which both cell type and drug appear in training, but not together, addressing the sparsity of high-throughput screens. (2) Unprofiled drugs: predicting responses for compounds entirely absent from the training data (zero-shot learning), a capability essential for virtual screening of novel chemical entities.

Knowledge graph construction

We constructed MAP-KG, a large-scale biomedical knowledge graph that integrates structured and unstructured information spanning small-molecule drugs, genes and their mechanistic relationships. MAP-KG aggregates curated evidence from 14 public databases (Supplementary Section A.10), encompassing drug–target interactions, gene functional annotations, pathway memberships and molecular mechanisms. By explicitly encoding biologically meaningful priors, MAP-KG provides a principled substrate for mechanism-aware modelling of cellular perturbation responses. As illustrated in Fig. 1a, the graph comprises two core entity types—drugs and genes—and two classes of relational edges: drug–gene and gene–gene interactions. We describe each component below and provide more representative examples in Supplementary Table 22.

Drug nodes. Each drug node corresponds to a small-molecule compound and is uniquely indexed using a PubChem Compound Identifier (PubChem ID)[39]. We selected PubChem for its extensive coverage of chemical space and robust cross-referencing with pharmacological and molecular biology resources. In addition to the PubChem ID, each drug is associated with two attribute types: (1) molecular structure (mol), encoded as SMILES strings; and (2) free-form textual knowledge (text), comprising the compound name, descriptions of MOA and biological therapeutic functions. Importantly, the MOA descriptions frequently reference target genes or proteins, thereby providing high-value semantic signals for modelling drug–gene relationships.

Gene nodes. Each gene node represents a human gene and is uniquely indexed by an Ensembl Gene Identifier (Ensembl ID)[40]. Ensembl IDs were chosen for their comprehensive coverage and stable, well-maintained mappings to alternative nomenclatures (for example, HGNC[41]). Each gene node is further associated with two attribute types: (1) the amino acid sequence (seq); (2) free-form textual knowledge (text), spanning the gene symbol, molecular or cellular functional annotations, concise gene summaries and descriptions of participating biological pathways. Together, these attributes capture complementary structural, functional and contextual information about each gene.

Relation edges. Edges in MAP-KG encode biologically meaningful relationships between entities. Rather than constraining relations to a fixed ontology, we annotate each edge with a natural-language textual description, preserving semantic nuance and enabling flexible representation learning. The graph includes two principal categories of relations. (1) Drug–gene relations, which capture mechanistic interactions such as binding, inhibition, activation or transcriptional regulation. Each edge is defined by a PubChem ID for the drug node, an Ensembl ID for the gene node, a free-text description of the interaction, and an optional quantitative score reflecting interaction strength or confidence. (2) Gene–gene relations, which represent functional or physical associations, including regulatory interactions and protein–protein interactions. Similarly, each edge is identified by a pair of Ensembl IDs and annotated with descriptive text and optional quantitative scores. All edges in MAP-KG are directed, reflecting the asymmetric nature of biological relationships (for example, regulator and target).

After normalization and deduplication, MAP-KG comprises 187,089 drug nodes, 22,924 gene nodes, 428,192 drug–gene edges and 266,054 gene–gene edges. This graph offers a compact yet expressive representation of drug mechanisms and gene interactions, forming a biologically grounded foundation for learning knowledge-enhanced representations of cellular perturbations. Unlike previous efforts[21,38,42,43], MAP-KG focuses on mechanistic knowledge linking genes and drugs: it integrates multimodal evidence and bridges modalities through rich textual descriptions, as well as assembling a large-scale knowledge graph from diverse knowledge sources.

Knowledge-driven multimodal pretraining

To learn mechanism-aware representations, we pretrain the encoders on MAP-KG by aligning heterogeneous entity attributes and relational context in a unified embedding space (Extended Data Fig. 1).

Intra-node knowledge encoding. As described in the 'Knowledge graph construction' section, each entity in MAP-KG is associated with intra-node knowledge from multiple modalities: (1) text, including entity names and natural-language descriptions (for example, MOA and functional annotations); (2) mol, the molecular structure of drug nodes represented as SMILES strings; and (3) seq, the canonical amino acid sequence of gene nodes.

For each modality, we adopt a modality-specific encoder (Φ_{text}, Φ_{mol} and Φ_{seq}) for knowledge encoding, where Φ_{text} is a biomedical language model (BioBERT[44]), Φ_{mol} is a molecule encoder for SMILES strings based on MoleculeSTM[37] and Φ_{seq} is a protein language model based on ESM-2 (ref. 45). Specifically, we freeze the weights of ESM-2 and MoleculeSTM to prevent feature degradation, and append four trainable multilayer perceptron layers to each of them for feature adaptation. Let $\mathcal{E}$ denote the set of all entities (drugs and genes) in the graph, and each entity $e_i \in \mathcal{E}$ denotes one of its modality-specific attributes as $k_i \in \{k_{\text{text}}, k_{\text{mol}}, k_{\text{seq}}\}$:

$$\mathbf{z}_{\text{text}} = \Phi_{\text{text}}(k_{\text{text}}), \quad \mathbf{z}_{\text{mol}} = \Phi_{\text{mol}}(k_{\text{mol}}), \quad \mathbf{z}_{\text{seq}} = \Phi_{\text{seq}}(k_{\text{seq}}), \tag{2}$$

where the encoders project heterogeneous biological inputs into a d-dimensional vector, which will be aligned later via contrastive pretraining. For simplicity, we use the name-text (that is, compound name or gene symbol) embedding from Φ_{text} as the canonical representation for e_i, denoted as $\mathbf{z}_i$.

Inter-node relation encoding. Inter-node relational knowledge in MAP-KG is expressed as natural-language descriptions on directed edges. For an edge from e_i to e_j with relation text r_{ij} (for example, 'inhibits'), we encode the relation using the text encoder:

$$\mathbf{z}_{ij} = \Phi_{\text{text}}(r_{ij}), \qquad \mathbf{z}_{ij} \in \mathbb{R}^d. \tag{3}$$

To explicitly model how an entity participates in a specific relation, we construct relation-conditioned entity embeddings. Specifically, we introduce a relation-conditioned composition function $\phi_{\text{rc}}(\cdot)$, implemented as a multilayer perceptron, which maps an ordered pair of embeddings to a d-dimensional representation:

$$\mathbf{z}_{i\to j} = \phi_{\text{rc}}([\mathbf{z}_i, \mathbf{z}_{ij}]), \qquad \mathbf{z}_{j\leftarrow i} = \phi_{\text{rc}}([\mathbf{z}_{ij}, \mathbf{z}_j]), \tag{4}$$

where $[\cdot,\cdot]$ denotes concatenation; $\mathbf{z}_i$ and $\mathbf{z}_j$ are canonical representations of e_i and e_j, respectively; $\mathbf{z}_{i\to j}$ denotes the embedding of entity e_i conditioned on its relation to e_j; and $\mathbf{z}_{j\leftarrow i}$ denotes the embedding of e_j conditioned on the relation from e_i. This order-sensitive composition function enables the model to distinguish the source entity from the target entity in directed relations. For instance, given the triplet (metformin, an inhibitor on the protein target, MT-ND1) in Fig. 1a, $\mathbf{z}_{i\to j}$ encodes metformin's active role as the regulator exerting inhibition, whereas $\mathbf{z}_{j\leftarrow i}$ characterizes MT-ND1 in the passive state of being targeted. This design captures the inherent directionality of biological interactions, effectively modelling the causal roles encoded in the graph edges.

Contrastive pretraining. As shown in Extended Data Fig. 1, we train the knowledge encoders with contrastive learning. Given a mini-batch $\mathcal{B}$, for each entity e_i, we first construct a pair of embeddings $(\mathbf{z}_i, \mathbf{z}'_i)$ that encode a specific biological prior and should be aligned in the representation geometry. Here $\mathbf{z}_i$ is the canonical representation, that is, the name-text embedding of e_i, and $\mathbf{z}'_i$ is randomly sampled according to two supervision strategies. (1) Intra-node attribute-level knowledge alignment (Extended Data Fig. 1a) pulls the attributes of each entity towards its name in the embedding space. In this case, $\mathbf{z}'_i$ is instantiated from $\{\mathbf{z}_{\text{text}}, \mathbf{z}_{\text{mol}}, \mathbf{z}_{\text{seq}}\}$, derived in equation (2). (2) Inter-node relation-level knowledge alignment (Extended Data Fig. 1b) pulls the relation-conditioned entity towards the linked neighbour in the embedding space. Here $\mathbf{z}'_i$ is instantiated as $\mathbf{z}_{j\leftarrow i}$, given a directed relationship from entity e_j, or $\mathbf{z}_{j\to i}$, given a directed relationship towards entity e_j, following equation (4). Then, we minimize the InfoNCE loss[46] over the mini-batch as

$$\mathcal{L} = -\frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}}\left[\log\left[\frac{\exp(\text{sim}(\mathbf{z}_i,\mathbf{z}'_i)/\tau)}{\sum_{k\in\mathcal{B}}\exp(\text{sim}(\mathbf{z}_i,\mathbf{z}'_k)/\tau)}\right] + \log\left[\frac{\exp(\text{sim}(\mathbf{z}_i,\mathbf{z}'_i)/\tau)}{\sum_{k\in\mathcal{B}}\exp(\text{sim}(\mathbf{z}_k,\mathbf{z}'_i)/\tau)}\right]\right] \tag{5}$$

where τ is a temperature hyperparameter.

The proposed knowledge-driven multimodal pretraining embeds drugs and genes in a shared space by aligning heterogeneous and fragmented biomedical evidence such that mechanistically or functionally similar drugs form coherent neighbourhoods in the representation geometry, rather than being isolated. Consequently, the geometry provides a transferable inductive bias: even when downstream perturbation data are sparse and noisy, the model can generalize by propagating information through these similarity-aware embeddings.

Knowledge-guided perturbation response prediction

In this section, we describe how knowledge-informed drug and gene representations are incorporated with a pretrained single-cell foundation model to guide perturbation response prediction (Extended Data Fig. 2). Formally, given a bulk $X \in \mathbb{R}^{B\times G}$ of B cells under the control condition, for each cell $x \in \mathbb{R}^G$ in the bulk, we first encode the unperturbed cellular state using a pretrained foundation model (STATE SE-600M)[15]:

$$\mathbf{T}, \mathbf{h} = \Phi_{\text{scfm}}(x),\ \mathbf{T} \in \mathbb{R}^{G\times d}, \mathbf{h} \in \mathbb{R}^d \tag{6}$$

where $\mathbf{T}$ is the latent gene-level embedding and $\mathbf{h}$ is the cell-level embedding that captures the global unperturbed cellular state, taken as the output of the prepended `SPECIAL` token. Then, we enhance the gene embeddings $\mathbf{T}$ with knowledge-informed gene-level representations $\mathbf{z}_{\text{seq}}$ (see the 'Knowledge-driven multimodal pretraining' section), and fuse them via a multilayer perceptron ϕ_{fuse}:

$$\mathbf{T}_{\text{fuse}} = \phi_{\text{fuse}}([\mathbf{T}, \mathbf{Z}_{\text{seq}}]), \mathbf{T}_{\text{fuse}} \in \mathbb{R}^{G\times d} \tag{7}$$

where $\mathbf{Z}_{\text{seq}} \in \mathbb{R}^{G\times d}$ is obtained by stacking the knowledge-informed representations of all G genes, and $[\cdot,\cdot]$ denotes concatenation along the feature dimension.

For a given perturbation $p \in \mathcal{P}$, we take its molecular SMILES string, obtain the knowledge-informed representation $\mathbf{z}_{\text{mol}}$ using the pretrained Φ_{mol} (see the 'Knowledge-driven multimodal pretraining' section) and scale the representation by $\tilde{\mathbf{z}} = \delta \cdot \mathbf{z}_{\text{mol}}$, where $\delta \in R$ is the log-transformed dose value applied. A transformer encoder module with four self-attention layers is then adopted, enabling the drug token to attend adaptively to mechanistically related gene tokens and gene-level reasoning to condition on the cell state. It takes as input a sequence constructed by concatenating $\tilde{\mathbf{z}}$, $\mathbf{h}$ and $\mathbf{T}_{\text{fuse}}$:

$$\hat{\mathbf{h}}^{(p)} = \Phi_{\text{enc}}(\tilde{\mathbf{z}}, \mathbf{h}, \mathbf{T}_{\text{fuse}}), \quad \hat{\mathbf{h}}^{(p)} \in \mathbb{R}^d$$

drug embeddings — cell state embeddings — gene embeddings

Here $\hat{\mathbf{h}}^{(p)}$ corresponds to the output embedding of $\mathbf{h}$. The perturbed cell embeddings are then passed through a multilayer perceptron head ϕ_{dec} to decode the perturbed gene expression profile:

$$\hat{x}^{(p)} = \phi_{\text{dec}}(\hat{\mathbf{h}}^{(p)}), \hat{x}^{(p)} \in \mathbb{R}^G \tag{9}$$

The three inputs to Φ_{enc} capture complementary factors of a perturbation response. We validate each representation's contribution via ablation (Supplementary Section A.4 and Supplementary Table 17). For a bulk X, the above procedure is applied independently to each cell. Here $\hat{x}_i^{(p)}$ and $\hat{\mathbf{h}}_i^{(p)}$ denote the predicted expression and embedding for the ith cell under perturbation p, respectively. The bulk-level predictions $\hat{X}^{(p)}$ and $\hat{\mathbf{H}}^{(p)}$ are obtained by stacking these cell-wise outputs.

Training objective. Following previous work[15], we train the perturbation predictor using a dual-space supervision objective that matches both (1) the perturbed expression profile in data space and (2) the perturbed cellular embedding in the STATE latent space. Specifically, for each training tuple $(X, p, X^{(p)})$, we first obtain the reference embedding matrix $\mathbf{H}^{(p)}$ by encoding each authentic perturbed cell $x_i^{(p)}$ with the foundation model Φ_{scfm} and stacking the resulting cell-level embeddings $\mathbf{h}_i^{(p)}$.

The loss function is then formulated as

$$\mathcal{L}_{\text{pert}} = \mathcal{L}_{\text{emb}} + \lambda \mathcal{L}_{\text{expr}} = \frac{1}{d}\left\|\bar{\hat{\mathbf{h}}}^{(p)} - \bar{\mathbf{h}}^{(p)}\right\|_2^2 + \lambda \frac{1}{G}\left\|\bar{\hat{x}}^{(p)} - \bar{x}^{(p)}\right\|_2^2 \tag{10}$$

where λ balances the two terms. Also, $\bar{x}^{(p)}$, $\bar{\hat{x}}^{(p)}$, $\bar{\mathbf{h}}^{(p)}$ and $\bar{\hat{\mathbf{h}}}^{(p)}$ denote the cell-wise means within the bulk:

$$\bar{x}^{(p)} = \frac{1}{B}\sum_{i=1}^{B} x_i^{(p)}, \qquad \bar{\hat{x}}^{(p)} = \frac{1}{B}\sum_{i=1}^{B} \hat{x}_i^{(p)}, \tag{11}$$

$$\bar{\mathbf{h}}^{(p)} = \frac{1}{B}\sum_{i=1}^{B} \mathbf{h}_i^{(p)}, \qquad \bar{\hat{\mathbf{h}}}^{(p)} = \frac{1}{B}\sum_{i=1}^{B} \hat{\mathbf{h}}_i^{(p)}. \tag{12}$$

Here the embedding-level supervision distils the perturbation-induced shift captured by the foundation model, providing a stable training signal when expression-level supervision is noisy or sparse.

Since each drug is represented by its SMILES string, the model is naturally extensible to any novel compounds at inference time in a zero-shot manner, as long as their SMILES strings are available. Moreover, Φ_{scfm} can be seamlessly replaced by other pretrained single-cell foundation models, making the proposed method broadly applicable across different backbone choices. Put differently, improvements from the scaling data and single-cell foundation model are orthogonal to our contributions.

To demonstrate the robustness of our method to parameter choices, we conduct ablation studies on three datasets over several key model hyperparameters (Supplementary Section A.3 and Supplementary Tables 14–16). We also provide the implementation resource cost in Supplementary Section A.5 and Supplementary Tables 19 and 20.

Datasets

We evaluated MAP on three large-scale single-cell perturbation datasets that profile transcriptional responses to small-molecule drug treatments across diverse cellular contexts. All datasets consist of paired control and perturbed single-cell RNA-sequencing profiles and were processed using a unified preprocessing pipeline to ensure consistency across experiments. Across all datasets, vehicle-only conditions were treated as controls (dimethyl sulfoxide when available; otherwise, we used the dataset-provided vehicle), and all compound-treated conditions were regarded as perturbations.

Dataset descriptions. The Tahoe-100M dataset[23] is a giga-scale single-cell perturbation atlas spanning cancer models. It comprises 95.6 million quality-controlled single-cell RNA-sequencing profiles collected from 50 cancer cell lines, of which 47 were retained for downstream analysis due to sufficient coverage. The dataset includes 379 drugs and 1,135 drug–dose combinations, yielding 52,886 perturbation conditions defined by the combination of cell line, compound and dose (median, 1,287 cells per condition). To provide a proof-of-concept evaluation and maintain computational tractability, we restricted training and validation to six cell lines spanning diverse tissues and cancer types: glioblastoma (A-172), kidney cancer (A-498), NSCLC (A-549), liver cancer (Hep-G2/C3A), pancreatic cancer (PANC-1) and skin melanoma (SK-MEL-2). This selection ensures that the benchmark reflects heterogeneous cellular contexts rather than a single lineage. To evaluate zero-shot generalization to unseen cell line–drug combinations, we held out 5% of drugs for each cell line as test drugs, with non-overlapping test sets across cell lines. To assess zero-shot generalization to entirely unprofiled drugs, we additionally held out 5% of all drugs ($n = 16$) during training.

The OP3 dataset[24] is a single-cell perturbation benchmark based on peripheral blood mononuclear cells. It contains 240,059 cells with the expression measured over 18,211 genes under 147 perturbation conditions, including 144 compounds administered at 1 μM. Dimethyl sulfoxide (14.1 μM) serves as the negative control, and two compounds (dabrafenib and belinostat at 0.1 μM) are included as positive controls. Cells are annotated into six immune cell types ($CD4^+$ T, $CD8^+$ T, natural killer (NK), regulatory T, B and myeloid populations); for B cells and myeloid cells, only a subset of drug conditions is available. All six cell types were included in our experiments. Following the same splitting strategy as for Tahoe-100M, we held out 5% of drugs per cell type to evaluate generalization to unseen cell type–drug combinations, and an additional 5% of all drugs to assess generalization to unseen drugs.

The SciPlex3 dataset[2] is a single-cell chemical perturbation dataset spanning three human cancer cell lines: A-549 (lung adenocarcinoma), MCF7 (breast adenocarcinoma) and K562 (chronic myelogenous leukaemia). It contains 581,777 cells treated with 187 compounds, with expression quantified over 58,347 genes. All three cell lines were included in our experiments, and we adopted the same train–test splitting scheme as described for the Tahoe-100M dataset.

The ComboSciPlex dataset[25] is a drug combination dataset that measures single-cell transcriptional responses to double drug perturbations. The dataset contains 63,378 cells and 32 perturbation conditions (7 single-drug conditions and 25 two-drug conditions). All samples were included in our experiments in which we trained MAP on single-drug perturbations and evaluated on two-drug combinations under three increasingly challenging splits: (1) both drugs seen during training, (2) one drug seen and one unseen and (3) both drugs unseen.

Data preprocessing. Gene expression preprocessing. For single-cell foundation model inputs, we restricted the gene space to 19,790 human protein-coding genes defined by Ensembl annotations. Each cell was normalized to a total of 10,000 unique molecular identifier counts and log-transformed via $\log 1p$ to obtain normalized expression values. For each cell, we selected the top values of the $L = 2,048$ genes ranked by log-normalized expression to form a fixed-length gene token sequence as input to the foundation model. Expression magnitudes were encoded using the soft-binning strategy described in ref. 15. This preprocessing pipeline was applied consistently across all three datasets.

Molecular structure preprocessing. For each compound, we used RDKit[47] to standardize its SMILES string and generate the canonical SMILES representation, which served as input to the molecule encoder.

Prediction targets. We evaluated predictions on two complementary gene sets: HVGs and DEGs. HVGs, defined as genes exhibiting the largest expression variability across cells and capturing informative biological heterogeneity, were computed independently for each dataset using `scanpy.pp.highly_variable_genes` applied to the raw gene expression count matrix. Following STATE[15], we adopted the Seurat v3 flavour, which models gene-wise variance directly on the count data, to select the top-2,000 HVGs based on a dataset-level

global variance model. The same dataset-specific HVG feature space was used consistently for training and evaluation across all baseline methods and across both zero-shot generalization settings. Since all experiments are conducted within each dataset (no cross-dataset evaluation), HVG definitions remain consistent between training and testing. DEGs, defined as genes exhibiting significant expression changes between perturbed and control cells for a given condition, to identify top DEGs, we first use a Wilcoxon rank-sum test with FDR control (adjusted $P < 0.05$) and then select the top-50 DEGs by ranking genes according to the absolute perturbation-induced log[fold change]. For both HVGs and DEGs, we applied library-size normalization followed by a log[1p] transformation.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All datasets used in this study are publicly available. We obtained the Tahoe-100M dataset from Huggingface (https://huggingface.co/datasets/tahoebio/Tahoe-100M), the OP3 dataset from the NeurIPS competition (https://www.kaggle.com/competitions/open-problems-single-cell-perturbations/data) and the SciPlex3 and ComboSciPlex datasets from the GEO at https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSM4150378 and https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE206741, respectively. MAP-KG is available via Huggingface at https://huggingface.co/datasets/RainGate/MAP-KG and the detailed knowledge source is provided in Supplementary Table 21. Source data are provided with this paper.

Code availability

All the code and models are publicly accessible via GitHub at https://github.com/MAGIC-AI4Med/MAP and via Zenodo at https://doi.org/10.5281/zenodo.21053262 (ref. 48).

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Author contributions

All authors contributed to the conception and design of the study. J.F., Z.Z. and X.Z. led the computational algorithm design, and performed data acquisition, analysis and interpretation. M.L., J.C. and X.Q. contributed to the acquisition and curation of the knowledge-base data used in this study. B.F., J.Z., Y.W., Y.Z. and W.X. provided technical and infrastructure guidance. All authors contributed to the drafting and revising of the paper. All authors approved the final version for publication and agree to be accountable for all aspects of the work.

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Competing interests

The authors declare no competing interests.

Additional information

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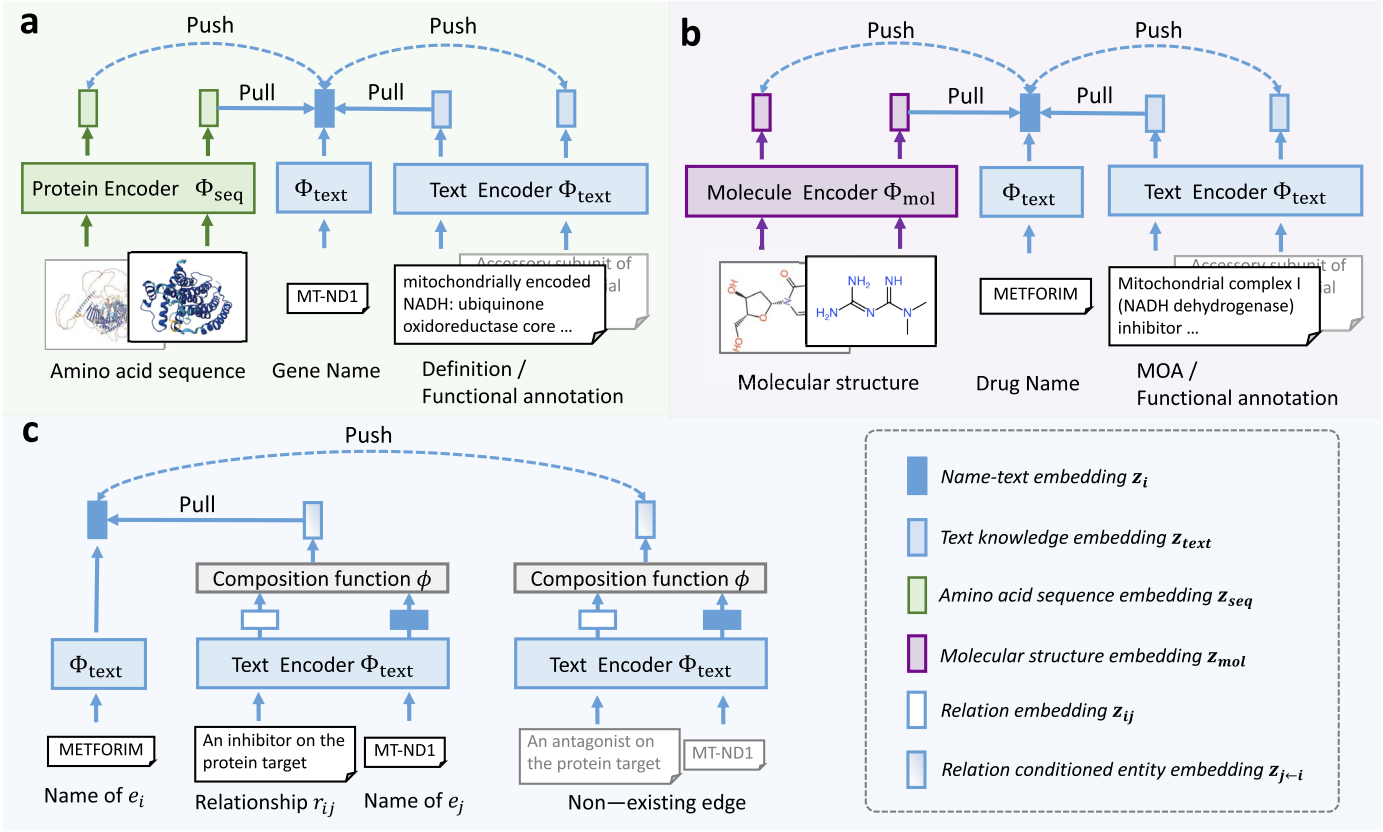
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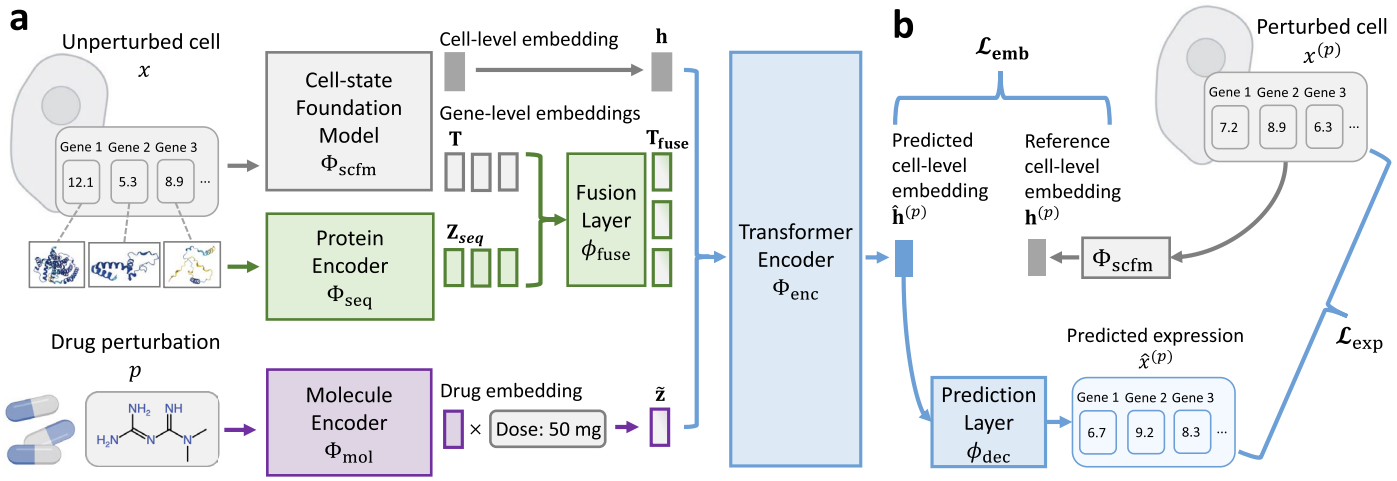
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Extended Data Fig. 1 | Knowledge-driven multimodal pre-training. We use modality-specific encoders to map heterogeneous biological input into the same vector space, and perform contrastive pre-training to align them, according to the biological priors. (**a**) and (**b**) show the inter-node attribute-level knowledge encoding and alignment for drug and gene entities. (**c**) shows the intra-node relation-level knowledge encoding and alignment between entities. Illustration in **b** created in BioRender; Zhou, X. https://biorender.com/rxukvvg (2026).

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Extended Data Fig. 2 | Knowledge-guided perturbation response prediction. (**a**) We devise a framework that seamlessly integrates the knowledge-informed drug and gene representations with a cell-state foundation model, to infer the perturbation response. (**b**) It is trained with both expression-level and embedding-level supervision to alleviate the measurement noise. Illustrations in **a** and **b** created in BioRender; Zhou, X. https://biorender.com/rxukvvg (2026).

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Data collection	Data collection and preparation scripts were implemented in Python 3.8.19.
Data analysis	The whole framework was implemented by Python 3.8.19. The code is available at https://github.com/MAGIC-AI4Med/MAP. Other data analysis softwares include: scanpy 1.9.8 and anndata 0.9.2.

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Population characteristics	The analyses were conducted using previously published and publicly available single-cell perturbation datasets. These datasets consist of gene expression measurements from cultured human cell lines and experimental cell systems subjected to chemical or genetic perturbations. The datasets do not contain identifiable information about individual human participants. Population-level demographic characteristics are not available in the datasets used in this study.
Recruitment	No participants were recruited for this study. All analyses were performed using previously published datasets obtained from public repositories. Details regarding participant recruitment procedures for the original studies can be found in the corresponding primary publications describing those datasets.
Ethics oversight	This study used only publicly available and de-identified datasets. Ethical approval was obtained in the original studies that generated these datasets, as described in the corresponding publications. No additional ethical approval was required for the present study.

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Data exclusions	No samples were excluded from the single-cell perturbation datasets beyond the standard preprocessing and quality-control procedures described in the Methods section. During MAP-KG construction, knowledge entries were excluded only when they failed to satisfy the minimum requirements for graph inclusion, for example because key entities necessary to define valid relations were missing. Redundant entities across knowledge sources were further removed through harmonization using standardized identifiers.
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